

EcoBirdNet – Bird Species Classifier

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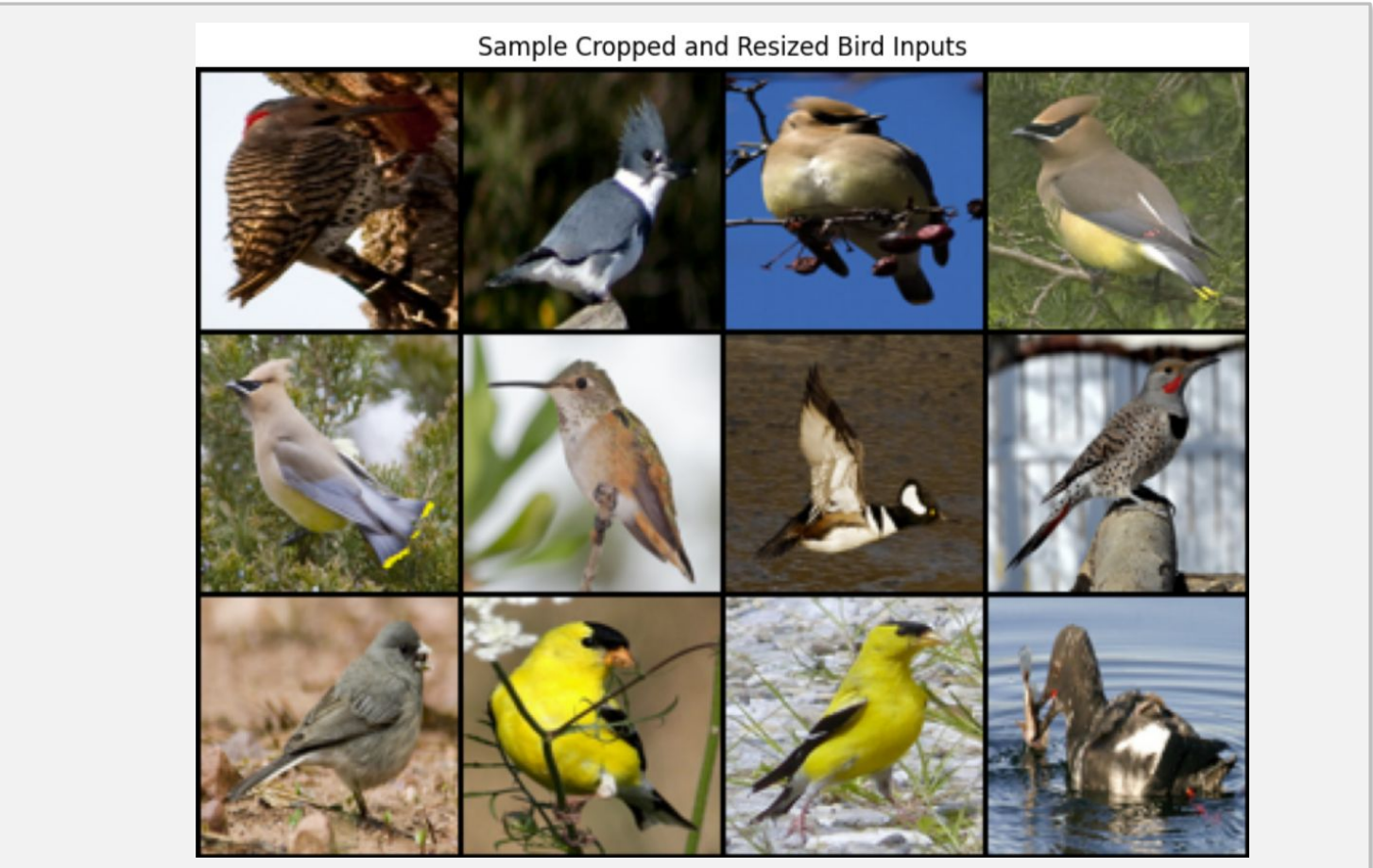
PHYS 417 - Advisor: Shih-Chieh Hsu; TAs: Aaron Hoyt, Jerry Li; Team Members: Daniel Michel, Rasool Ray, Arthur Wang

Motivation

- EcoBirdNet classifies bird species from cropped RGB photographs.
- The project adapts a CNN image-classification workflow for a small 16-class bird dataset.
- The goal is a clear, lightweight classifier with slide-ready evaluation outputs.

Dataset & Preprocessing

- Inputs: X.npy, y.npy, class_names.npy
- Images are cropped, resized, clipped, normalized, and converted into $96 \times 96 \times 3$ tensors.
- Train / validation / test splits are used, with training augmentation to improve generalization.



Research Question

- RQ1:** How accurately can a compact CNN classify 16 bird species from a limited, preprocessed image dataset?
- RQ2:** How does classification performance vary across species, and which birds are most frequently confused due to visual similarity?
- RQ3:** To what extent can the model's predictions be interpreted by users in a bird-identification context?

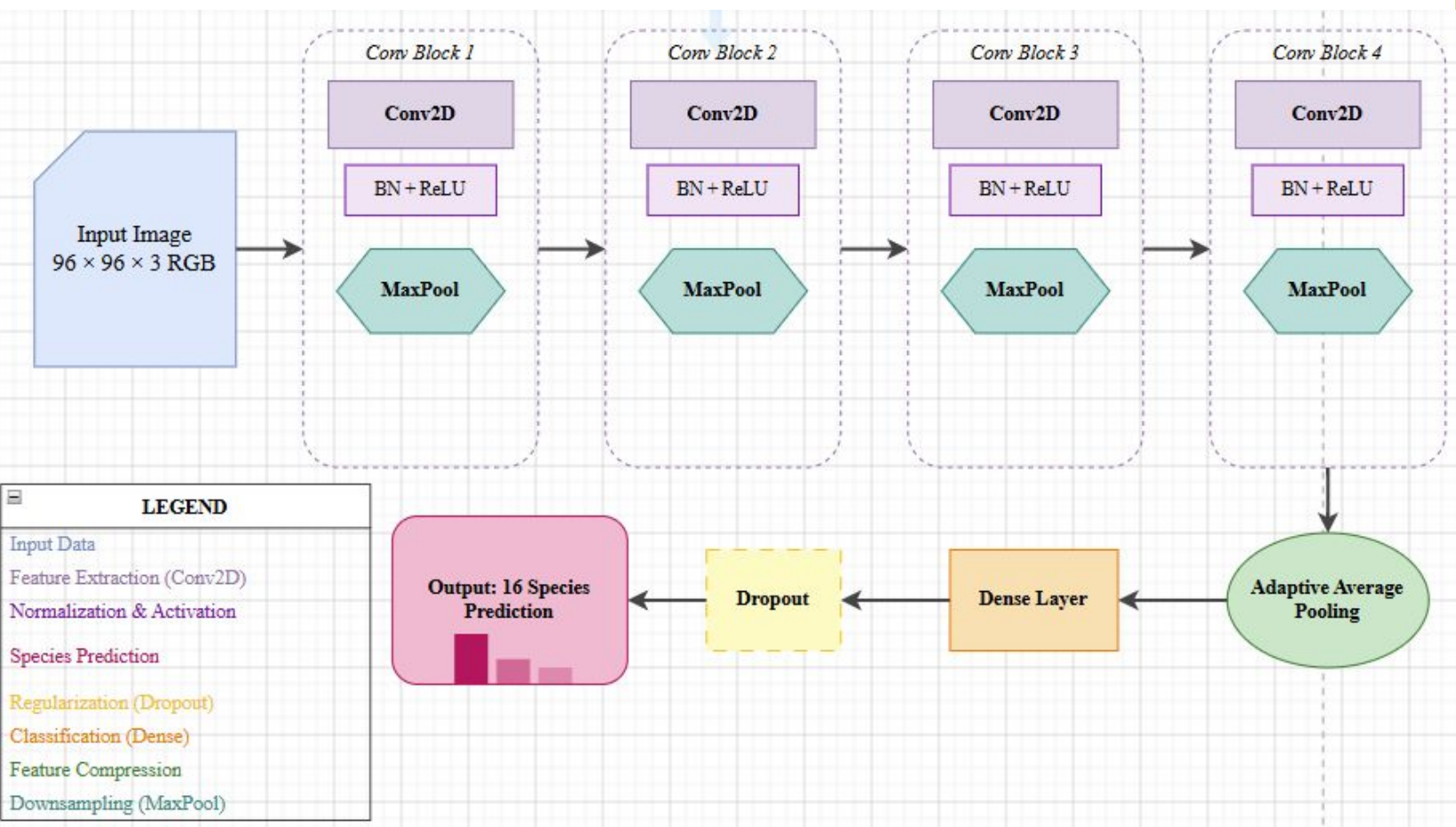
Poster Takeaway

EcoBirdNet provides a compact baseline for recognizing 16 bird species.

The model learned useful visual features, but generalization to unseen images is still limited.

It performed better on visually distinct species and struggled with similar or variable classes.

Approach



Model Architecture

Input: 96×96 RGB image
 $4 \times$ Conv2D $\rightarrow$ BatchNorm $\rightarrow$ ReLU $\rightarrow$ MaxPool blocks
Adaptive average pooling $\rightarrow$ dense layer $\rightarrow$ dropout $\rightarrow$ 16-class output

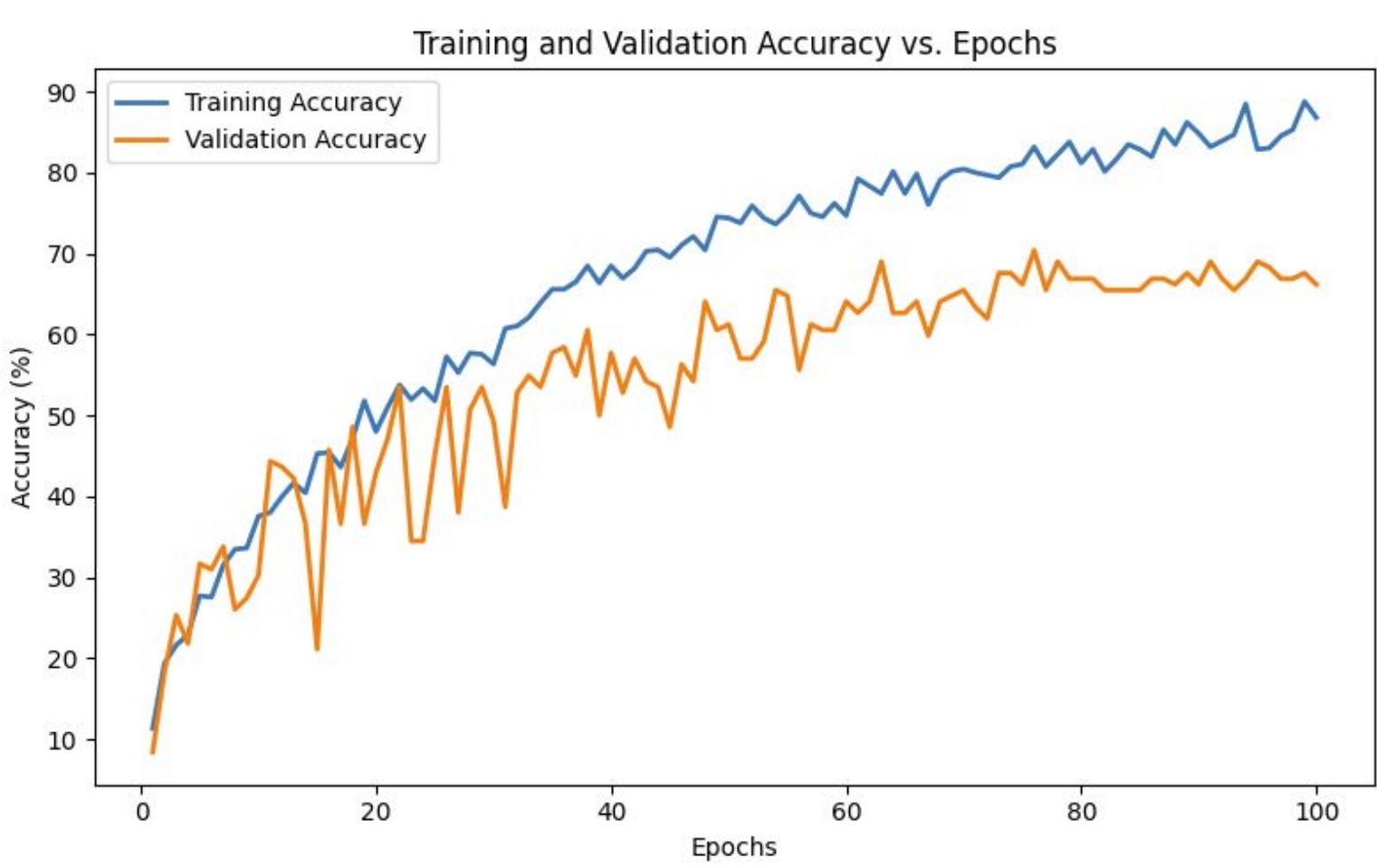
Our model is a CNN adapted from Lab 4's image-classification pipeline. It takes a $96 \times 96 \times 3$ RGB image as input and applies four convolutional blocks:

Conv2D $\rightarrow$ BatchNorm $\rightarrow$ ReLU $\rightarrow$ MaxPool. These blocks learn important bird features such as color, texture, and body shape. The model then uses adaptive average pooling, followed by a dense layer and dropout. The final layer outputs predictions for 16 bird species.

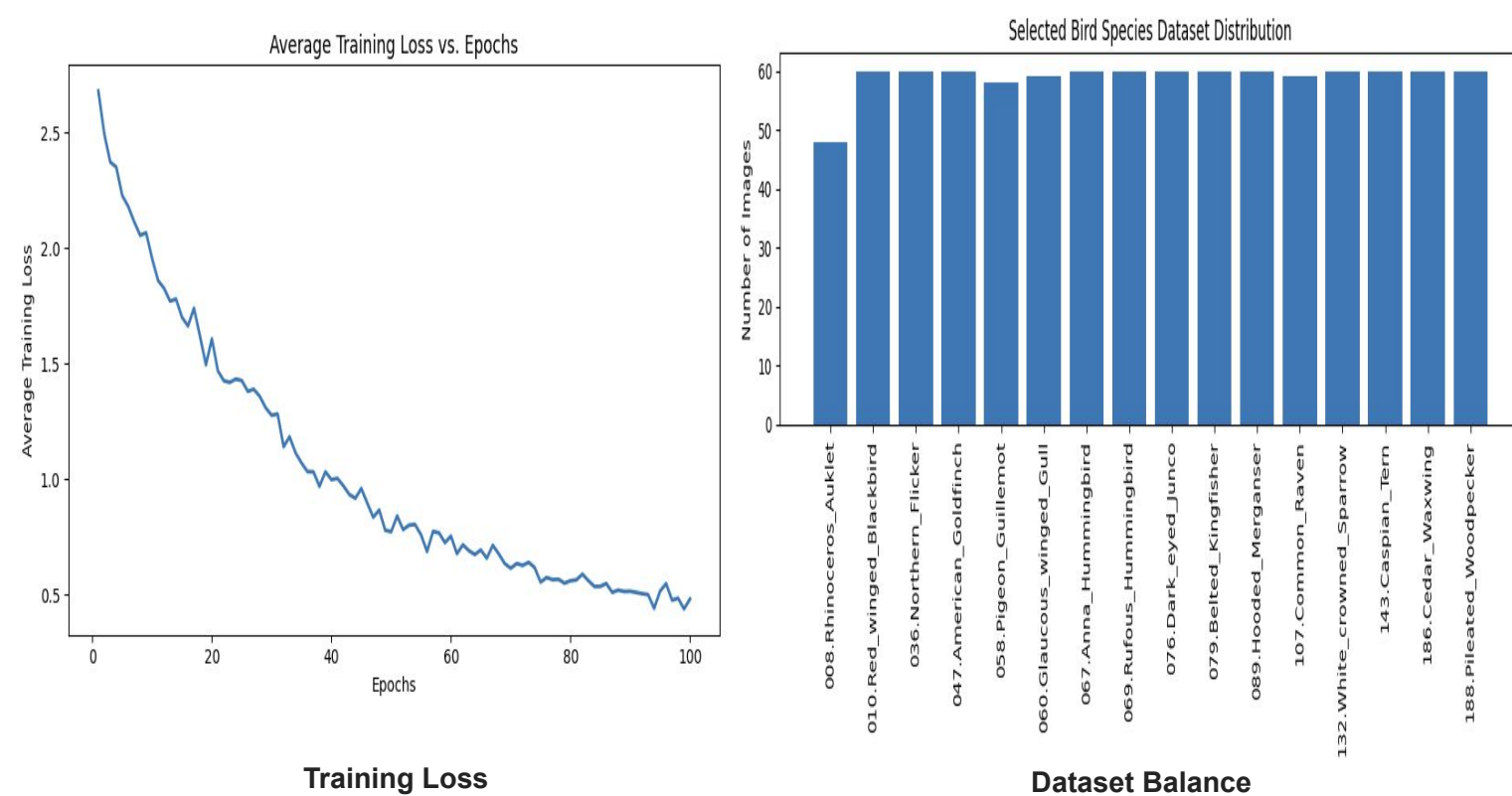
Training Setup

- Optimizer: Adam
- Learning rate: 0.001
- Epochs: 100
- Loss: weighted cross-entropy for class imbalance
- Scheduler: ReduceLROnPlateau
- Best model saved using validation accuracy

Evaluation Metrics



Class-Level Performance



Main Results

Final results after deploying model:

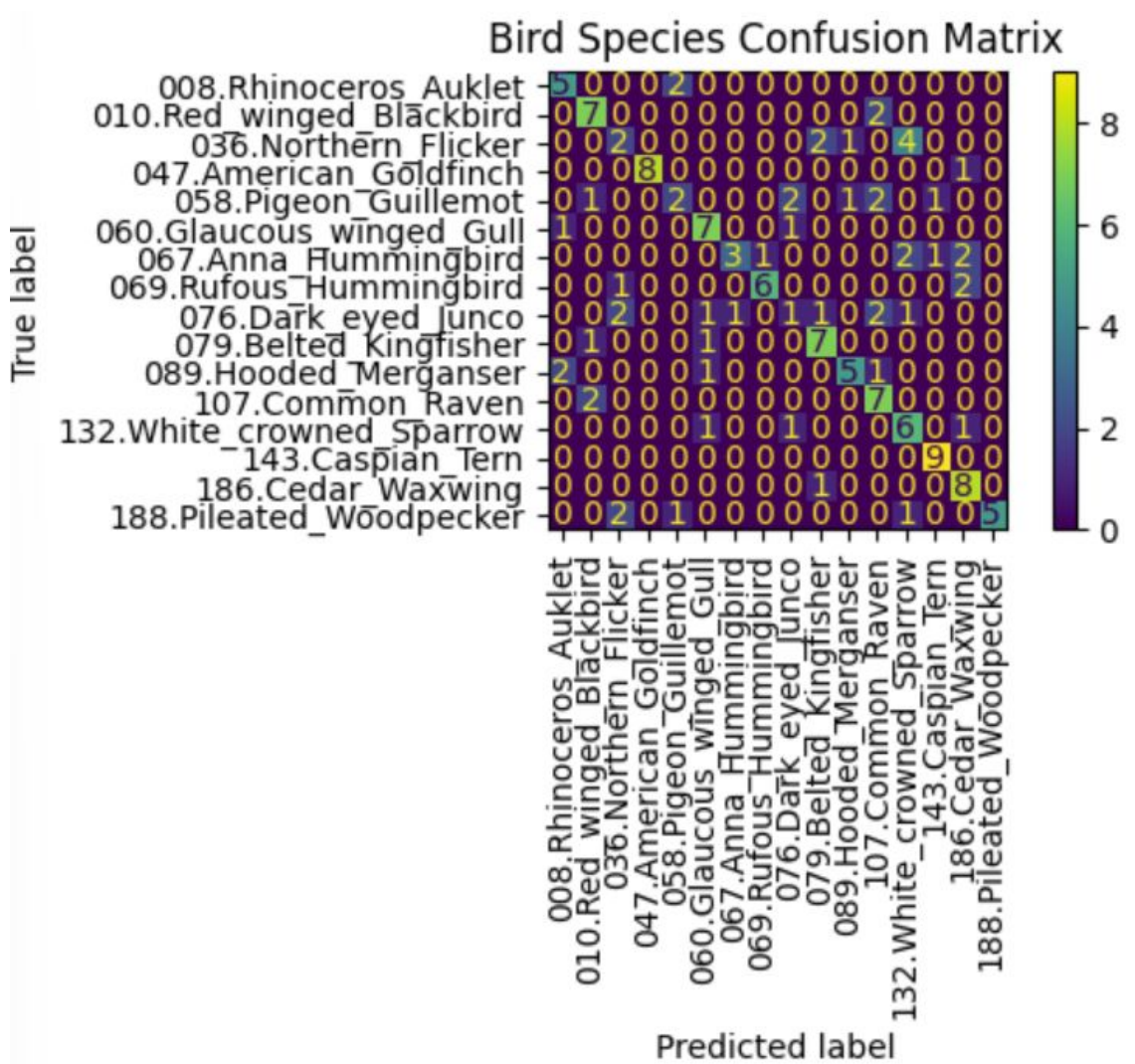
- Best validation accuracy: 73.94%
- Test accuracy: 62.68%
- Strongest / weakest species: Caspian_Tern (100.0%) / Dark_eyed_Junco (22.2%)

Interpretation:

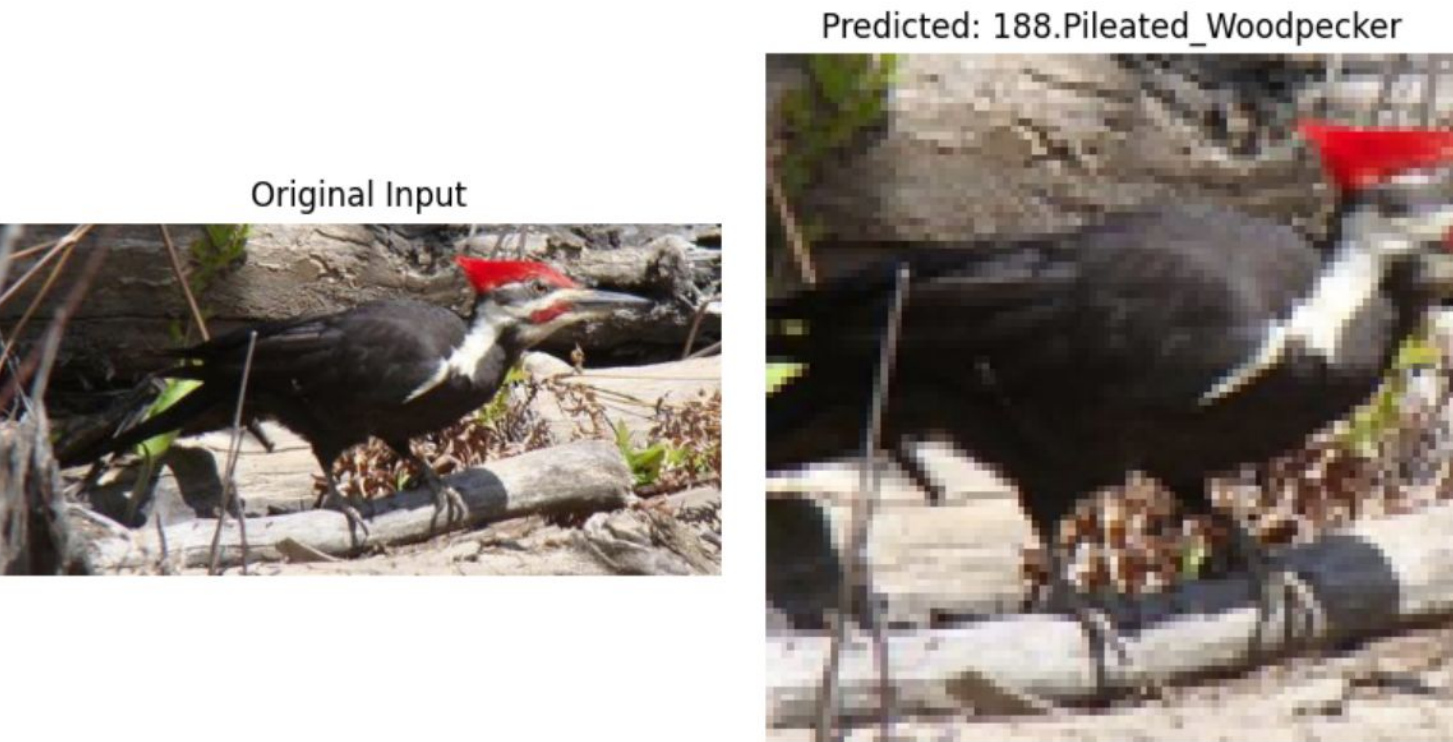
The compact CNN reached 73.94% best validation accuracy and 62.68% test accuracy. It learned useful visual features, but generalized less well to unseen images. This gap suggests moderate overfitting, a common issue in fine-grained bird classification.

Class accuracy varied widely. Caspian Tern was highest at 100.0%, while Pigeon Guillemot and Dark-eyed Junco were lowest at 22.2%. The model worked better on visually distinct species, but struggled with visually similar or more variable classes.

Results Visuals



Example Identification



True species: 188.Pileated_Woodpecker
Predicted species: 188.Pileated_Woodpecker
Confidence: 78.00%

Try out the app for yourself!

